

508 paid FTE per month to meet service level

THE ANSWER — JULY TO SEPTEMBER

508 paid FTE

per month to hold **80% / 20s** phone and **80% / 60s** chat at the June run-rate **+10%** seasonal uplift.

PRODUCTIVE HOURS

68,827

per month — the BPO bill

CONTACTS

294k

all languages, all channels

SCENARIO RANGE

462–579

LOW flat → HIGH trend

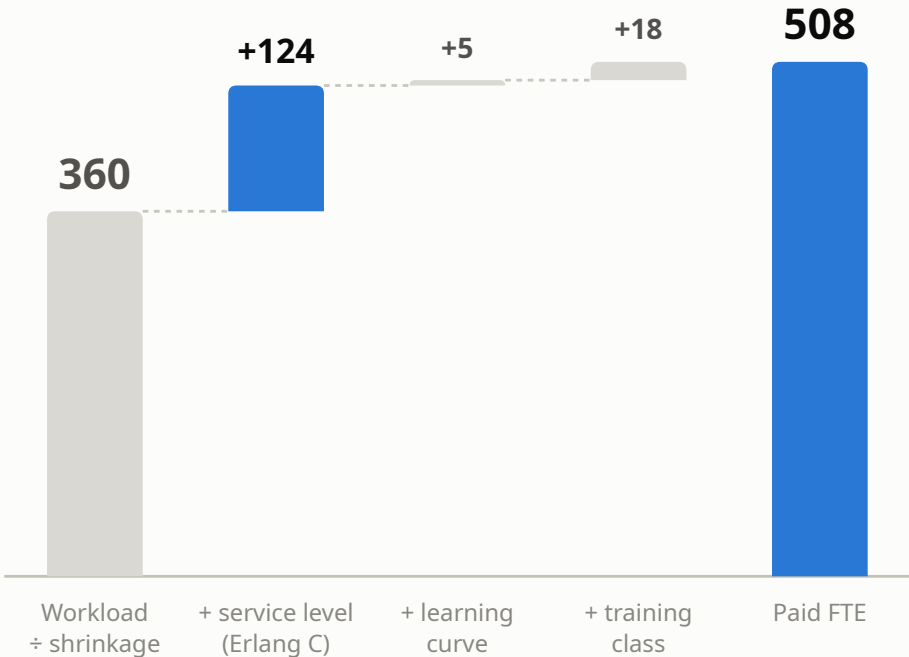
P10–P90

425–606

$\sigma = 13.8\%$

A QUARTER OF THE REQUIREMENT IS SERVICE LEVEL, NOT WORKLOAD

+34% — this is the brief's requirement



A workload model returns **360 FTE** and **misses SLA** — it implicitly assumes 100% occupancy. Erlang C says English inbound runs at **70%**; the idle 30% is the 20-second answer time.

TWO LEVERS, BOTH FROM THE SAME EVIDENCE

Pool the four non-English queues –51 FTE

German phone runs at 42% occupancy, Italian at 28% — four small queues each paying the 80/20 minimum-seat floor separately. All four are already in BPO 3.

Validate chat concurrency –75 FTE

Chat is 39% of the whole operation. The given 1.2 is far below the industry 2–3 and repeats on every row — it reads as a placeholder, not a measurement.

One number needs an owner ±78 FTE

The workbook states English AHT twice — **908.8s** by reason, **709.8s** by channel. Run both ways the answer is **508 vs 585**. That is 15% of the plan, unresolvable without the business.

How the number is built — and what the data would not tell us

THE CHAIN

contacts × AHT	workload hours
phone & chat → Erlang C	concurrent seats
email & outbound ÷ occupancy	seat-equivalent hours
÷ (1 – 18% shrinkage)	paid hours
÷ 173.33 h per FTE	FTE meeting SLA
÷ tenure efficiency	on-floor FTE
+ training class	paid FTE

Erlang C returns seats, not FTE. 11 concurrent seats × 730 h ÷ 0.82 ÷ 173.33 = **56.5 FTE**, against 39.5 from workload alone on the same queue.

Monthly-average arrival rates make this a *floor*: real intraday peaks and overnight minimum-staffing push interval-level staffing up, never down.

Everything here is reproducible. One command regenerates the whole plan from the workbook; a second re-verifies it with 72 independent checks, including Erlang C against a discrete-event simulation and the full July chain re-derived from raw cells through a separate code path. Appendix A5 has the detail.

DATA QUALITY — 12 CHECKS, EVERY DECISION LOGGED

WARN	channel rows sum to language total	17/30 language-months disagree with their own 'Tot' row; max 2 contacts...
PASS	EN CSAT reason tickets tie to Contact Volume English total	max ratio - 1 = 0.00e+00 across 6 months (ratios 1.000000, 1.000000...
FIX	CSAT reported against zero tickets	3 reason-months carry a CSAT score with 0 tickets (e.g. 18.2. Positive...
FIX	reason label missing	one reason row has a blank label carrying 3,729 tickets (0.63% of English...
FIX	AHT outliers vs reason median	1 reason-months outside 0.5x-2.0x their own median; worst: 1.4. Pricing...
WARN	channel pair Inbound == Outbound	identical in 29/30 language-months
WARN	channel pair Email == Chat	identical in 23/30 language-months

The validation that passed matters most: the per-reason ticket counts tie to the English contact volume in all six months at a ratio of **exactly 1.000000** — so the two sheets describe the same population and the reason-mix join is safe.

SLA attainment has fallen 95% → 92% while volume doubled (February reports 110%, which is impossible). That is the business case for this transformation, sitting unused in the source data.

TWO OPEN QUESTIONS, PRICED

1 · Which English AHT is authoritative?

Reason-level **908.8s** vs channel-level **709.8s**. Both cannot be right — 25% of English contacts are outbound at 107s. **Worth 78 FTE (15%)**.

2 · Is chat concurrency measured or assumed?

1.2 on every BPO and every language, under a header that says "English". **Worth 75 FTE**.

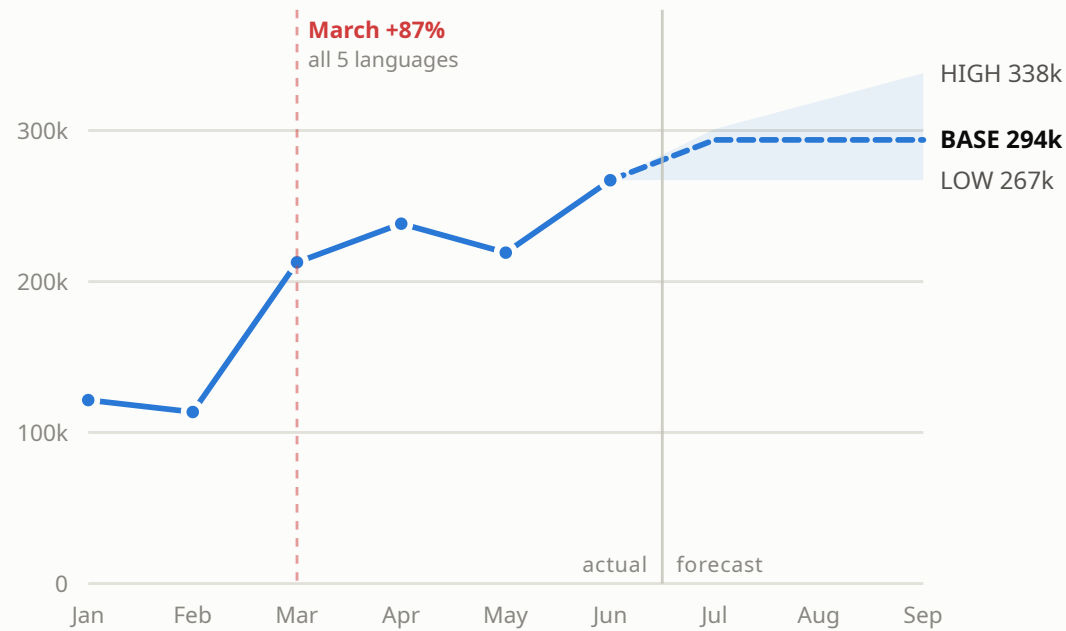
ALSO REQUESTED

- 24+ months of history — 6 cannot support a seasonal model
- 15/30-minute arrival data — monthly Erlang is a floor
- What changed in March — system, market, or demand?
- Cost per productive hour by BPO — to rank levers in €
- True outbound volumes — the mirrored columns cannot be real

Structural finding: 45 reason codes collapse to **25** distinct AHT profiles, and **6** of those clusters cross reason families — AHT is measured on a different taxonomy than the reason codes, and neither nests inside the other.

Forecast, capacity and the range around it

VOLUME — SIX MONTHS OF ACTUALS, THREE OF FORECAST



The baseline is June, not an average. March is a **structural break**, not seasonality — all five languages step together (1.60×–2.06×) across every channel and all 49 reason codes. Consumer demand does not double in lockstep across five independent markets; a reporting or scope change does. Either way the action is the same: anchor on post-break months only.

BASE is held flat across Jul–Sep — not a claim that demand is flat, but a refusal to invent a summer curve from six months containing a level shift. HIGH is the summer-peak case; slide 5 carries the trigger that switches to it.

CAPACITY BY LANGUAGE — BASE, PER MONTH

LANGUAGE	CONTACTS	PROD H	WORKLOAD FTE	SLA FTE
English	174,128	40,298	242	284
German	36,135	8,220	36	58
Spanish	32,879	7,133	31	50
French	26,662	6,773	26	48
Italian	24,066	6,402	25	45
Total	293,869	68,827	360	484

BY BPO — THE VENDOR BRIEF

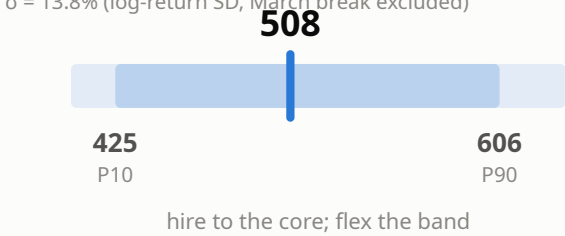
BPO	LANGUAGES	FTE	PROD H
BPO 3	DE IT ES FR	201	28,529
BPO 1	English 50%	142	20,149
BPO 2	English 50%	142	20,149

English AHT is the BPO1/BPO2 50/50 blend, so total capacity does not depend on how English is split between the two vendors.

SCENARIOS — PAID FTE

CASE	JUL	AUG	SEP	BASIS
LOW	462	462	462	June run-rate, no uplift
BASE	508	508	508	June × 1.10, held flat
HIGH	518	551	579	English on trend × 1.10

Paid FTE — 80% confidence band
σ = 13.8% (log-return SD, March break excluded)

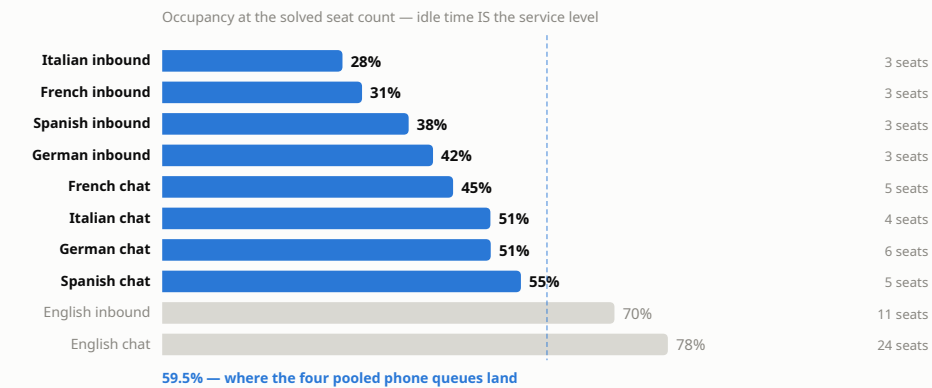


Do not hire to a point estimate. Contract a ~490 FTE core and hold a cross-trained flex pool of 60–100 against the P90, released monthly on the re-forecast trigger.

Backfill alone runs at 19 hires/month at 36% annual attrition — with a 4-week class and a 3-month ramp, recruitment must lead the forecast by a full quarter.

Two strategies the arithmetic supports — and one it kills

STRATEGY 1 · POOL THE FOUR NON-ENGLISH QUEUES **-51.4 FTE (10%)**



The occupancy column is the finding. Those queues are not idle by choice — 80/20 on a queue of 0.84 erlangs, staffed 24/7, needs 3 seats no matter how few calls arrive. **Four small queues pay that floor four times.**

QUEUE	SILOED	POOLED	SL	SAVED
Inbound DE IT ES FR	12 seats	7	86%	25.7
Chat DE IT ES FR	20 seats	14	86%	25.7
Total	≈ 8,900 productive hours off the BPO 3 bill			51.4

WHY IT WORKS, AND WHAT IT COSTS

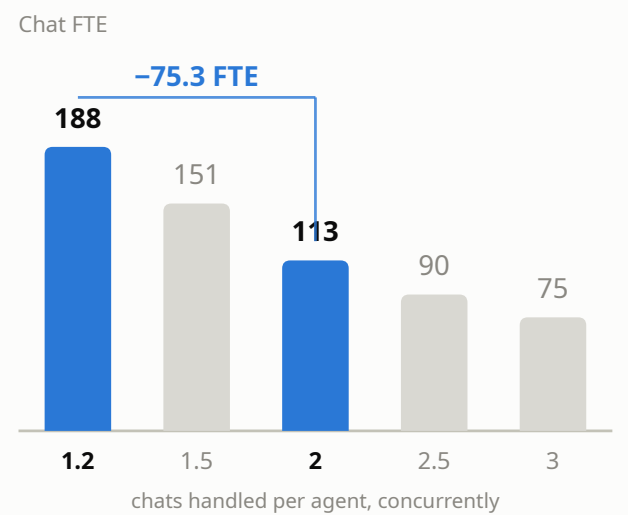
All four languages already sit in BPO 3, so this is a routing and skilling change, not a vendor change. Both pooled queues were **re-solved through the same Erlang engine** — the service level goes up, not down. Phase it: start with the overnight window, where the floors bite hardest.

Tested and rejected: deflect phone → chat

The intuitive lever. Modelled end-to-end it **costs** FTE, because English chat consumes **958 agent-seconds** per contact against phone's **458**.

SHIFT	PHONE	CHAT	FTE	Δ
0%	11	24	159.2	0.0
10%	10	26	162.6	+3.4
20%	9	28	166.1	+6.8
30%	9	30	174.6	+15.4

STRATEGY 2 · CHAT CONCURRENCY **-75.3 FTE (15%)**



Chat is **188 FTE — 39% of the entire operation**, the largest single line in the plan. The given 1.2 sits under a column headed "Chat concurrency *English*" yet repeats on every BPO and every language.

Ask whether it was measured. If it was, the tooling or the routing rules are the constraint and there is a clear case to fix them. If it is a placeholder, the plan carries a 75 FTE error. Either answer is worth having.

Where the real prize is. The top three handling-hour drivers — **1.1 Availability** (5,026 h), **3.3 Meeting point** (4,712 h) and **3.1 Voucher not received** (3,844 h, CSAT 65%) — are high-volume, low-CSAT and automatable. That is a product ask, not a scheduling change, so it sits beside these two rather than among them.

Leading the transformation — and how AI was actually used

2A · 30 / 60 / 90

0–30 · Decide, then build

Four decisions gate everything downstream — none are technical: **who owns the AHT definition** (the ± 78 FTE question is unresolvable without an owner); **forecast granularity** (language $\times$ channel $\times$ interval — anything coarser cannot produce an SLA-compliant number); **who signs off assumptions** and how often; **where the model lives**. Baseline the manual model's accuracy in week 1 — without it there is nothing to beat.

31–60 · Shadow, don't switch

New model runs in parallel on English — 59% of volume, all of the growth, all of the forecast risk (σ 19.5% vs under 9% elsewhere). Weekly reconciliation against the manual plan. The manual plan stays *the* plan.

61–90 · Cut over, then extend

English cuts over once it has beaten the baseline for six consecutive weeks. DE/IT/ES/FR follow. Escalation and override rules written down before, not after.

"Done" at 90 days: the model produces the weekly plan unaided, beats the manual baseline on MAPE for six straight weeks, the three specialists run it without me, and every assumption is versioned and dated.

2B · BRINGING THREE SPECIALISTS ALONG

Assess with an artefact, not an opinion. Give all three the same task on this dataset — clean it, forecast one language, document the assumptions. What comes back places them on forecasting depth $\times$ AI fluency far better than a conversation does.

Set the expectation explicitly: AI is the tool, they own the judgement. The two things AI got wrong in this exercise were both plausible-sounding domain defaults — exactly what an expert reviewer catches and a novice ships.

Protect BAU: six weeks of shadow running, paired shadowing, and a written prompt playbook. Capability is built *alongside* the plan, never instead of it.

2C · WHAT I WOULD MEASURE

Leading — inputs with a named owner and a dated assumption · time to produce a plan · % of runs reproducible from source · open data-quality items (**starts at 12**) · specialists running it solo (0 $\rightarrow$ 3).

Lagging — forecast MAPE *and* BIAS by language · SLA attainment (baseline **92%, falling**) · FTE plan-vs-actual · shrinkage by bucket · cost per contact.

Lead with bias, not accuracy. Accuracy hides direction, and direction is what a workload-only model gets wrong — it is short by 34%, every single month.

1C · HOW AI WAS USED — AND WHERE IT WAS WRONG

WHERE	VERDICT	WHAT HAPPENED
Sheet parsing, anomaly sweep, reconciliation	ACCEPTED	Deterministic Python. I verified the cross-sheet reconciliation independently.
Erlang C implementation	ACCEPTED	After checking it by hand and against a simulation. Never ship a staffing formula you have not verified.
First-pass model structure	REJECTED	Produced workload $\div$ shrinkage <i>and</i> a separate Erlang table that never met. Meets the brief only once occupancy feeds the FTE.
"Deflect phone $\rightarrow$ chat"	REJECTED	Generic WFM instinct contradicted by this dataset's AHTs. Caught by modelling it, not by reading it.
Reason-taxonomy clustering	CHANGED	Concluded "AHT is family-level!"; testing showed clusters cross families in 6 of 11 groups.
Narrative drafting	CHANGED	Defaulted to hedging. An ops audience needs a number and a recommendation.

What I would do differently. State the acceptance test *before* generating — "the FTE number must satisfy the SLA constraint" — rather than reviewing after. AI is reliable at parsing, arithmetic and structure, and unreliable at knowing **which of two conflicting numbers in a workbook is the real one**. That gap is a judgement call requiring the business, and no amount of prompting closes it.

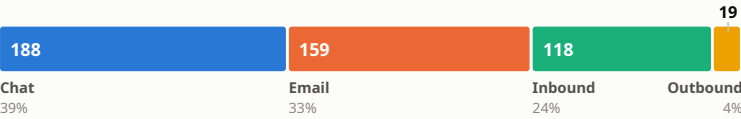
Tools: Claude (Opus 5) in Claude Code for the analysis, the Python model and this deck. The verification suite found **three real defects in my own model** — a growth test that flagged a declining language as growing, a HIGH scenario that could land below BASE, and a mis-reported trend slope — all fixed at source.

Every queue, end to end — BASE, July

The full working behind slide 3. Occupancy below 100% on the Erlang rows is the cost of the service level.

LANGUAGE	CHANNEL	CONTACTS	AHT S	MODEL	ERLANGS	SEATS	FTE-EQ SEATS	OCC	ACHIEVED SL	WORKLOAD H	PRODUCTIVE H	WORKLOAD FTE	SLA FTE
English	Chat	42,882	1,150	Erlang C	18.77	24	20.0	78%	86.2%	13,699	14,600	96.4	102.7
English	Email	42,882	1,150	deferred	—	—	22.1	85%	—	13,699	16,116	96.4	113.4
English	Inbound	44,182	458	Erlang C	7.69	11	11.0	70%	82.6%	5,615	8,030	39.5	56.5
English	Outbound	44,182	108	deferred	—	—	2.1	85%	—	1,319	1,552	9.3	10.9
French	Chat	6,566	900	Erlang C	2.25	5	4.2	45%	92.5%	1,641	3,042	11.5	21.4
French	Email	6,566	600	deferred	—	—	1.8	85%	—	1,094	1,287	7.7	9.1
French	Inbound	6,765	360	Erlang C	0.93	3	3.0	31%	93.3%	677	2,190	4.8	15.4
French	Outbound	6,765	115	deferred	—	—	0.3	85%	—	216	254	1.5	1.8
German	Chat	8,899	900	Erlang C	3.05	6	5.0	51%	91.3%	2,225	3,650	15.7	25.7
German	Email	8,899	700	deferred	—	—	2.8	85%	—	1,730	2,036	12.2	14.3
German	Inbound	9,168	360	Erlang C	1.26	3	3.0	42%	85.7%	917	2,190	6.5	15.4
German	Outbound	9,168	115	deferred	—	—	0.5	85%	—	293	345	2.1	2.4
Italian	Chat	5,927	900	Erlang C	2.03	4	3.3	51%	84.1%	1,482	2,433	10.4	17.1
Italian	Email	5,927	800	deferred	—	—	2.1	85%	—	1,317	1,549	9.3	10.9
Italian	Inbound	6,106	360	Erlang C	0.84	3	3.0	28%	94.8%	611	2,190	4.3	15.4
Italian	Outbound	6,106	115	deferred	—	—	0.3	85%	—	195	229	1.4	1.6
Spanish	Chat	8,097	900	Erlang C	2.77	5	4.2	55%	84.2%	2,024	3,042	14.2	21.4
Spanish	Email	8,097	600	deferred	—	—	2.2	85%	—	1,350	1,588	9.5	11.2
Spanish	Inbound	8,342	360	Erlang C	1.14	3	3.0	38%	88.7%	834	2,190	5.9	15.4
Spanish	Outbound	8,342	115	deferred	—	—	0.4	85%	—	266	314	1.9	2.2
Total		293,869								51,203	68,827	360.2	484.2

WHERE THE 484 FTE SITS



Chat is the single largest line at 39%, which is why concurrency is the biggest available lever.

Outbound is 4% despite carrying a quarter of all contacts — its AHT is 107s against email's 1,150s. That is also why the mirrored outbound column matters far less to capacity than it does to data credibility.

Email at 159 FTE is modelled as deferred work at 85% planned occupancy — an explicit assumption, not from the workbook. An 80%-in-120-minute target is close to real-time, so if the business treats email as a live queue rather than a backlog, this line grows.

The **deferred** rows carry no Erlang figures by design: a 120-minute target is a backlog commitment, not a queue discipline, so seats are derived from hours and planned occupancy instead.

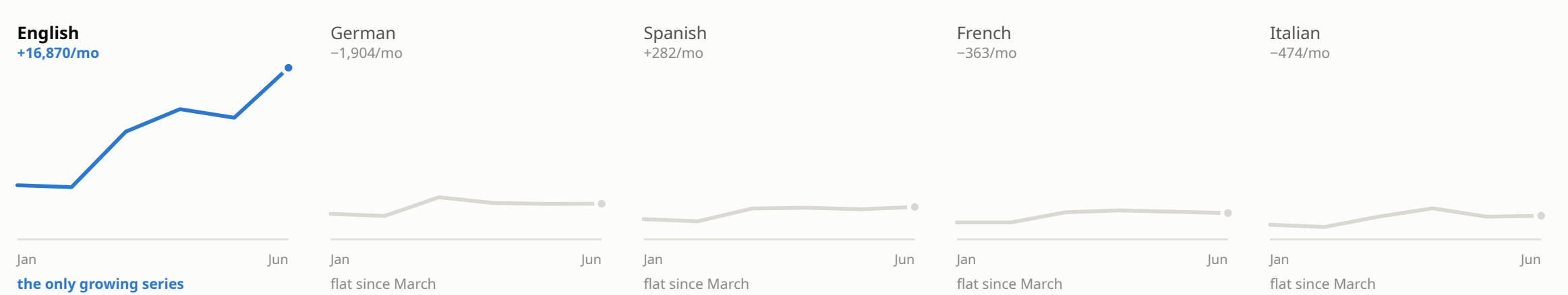
Data-quality log — all 12 checks

Condensed to five rows on slide 2. This is the full log, including the checks that passed.

#	STATUS	CHECK	FINDING	DECISION	CAPACITY IMPACT
1	WARN	channel rows sum to language total	17/30 language-months disagree with their own 'Tot' row; max 2 contacts (0.0120% of the total)	rounding residue in synthetic data. Channel rows are the source of truth (each has its own AHT and SLA); language totals are derived from them	<0.0120% of volume - immaterial to capacity
2	PASS	EN CSAT reason tickets tie to Contact Volume English total	max ratio - 1 = 0.00e+00 across 6 months (ratios 1.000000, 1.000000, 1.000000, 1.000000, 1.000000, 1.000000)	the two sheets describe the same population - safe to join	validates the reason-mix join used for AHT weighting
3	FIX	CSAT reported against zero tickets	3 reason-months carry a CSAT score with 0 tickets (e.g. 18.2. Positive feedback / February -> CSAT 88)	null the CSAT; keep the row so the ticket series stays complete	none - CSAT is not a capacity input
4	FIX	reason label missing	one reason row has a blank label carrying 3,729 tickets (0.63% of English volume)	relabel 'UNMAPPED' and KEEP the volume - it is real contact demand	0.63% of English volume would be lost if dropped
5	FIX	AHT outliers vs reason median	1 reason-months outside 0.5x-2.0x their own median; worst: 1.4. Pricing June = 271s vs median 1408s	replace with the reason's own median	<0.5% of the English weighted AHT
6	WARN	channel pair Inbound == Outbound	identical in 29/30 language-months	synthetic mirror - keep as-is (conservative), flag in assumptions	outbound volume is not credible; ~8 FTE over-stated if true outbound is ~10% of inbound (AHT 107s, so the exposure is small)
7	WARN	channel pair Email == Chat	identical in 23/30 language-months	synthetic mirror - keep as-is (conservative), flag in assumptions	channel split is fabricated, but email/chat AHT are close (1,150 vs 1,150 EN) so the capacity exposure is minimal
8	WARN	BPO x language routing constraint	9 of 15 rows violate 'Other information' B5/B6 (e.g. ('BPO 1', 'French'))	drop unroutable rows before any AHT lookup	none if dropped; wrong-language AHT if not
9	PASS	shrinkage components sum to Tot Shrink	max Tot Shrink - sum(components) = 0.00e+00 over 15 language-BPO pairs; Tot Shrink is 18%-18% everywhere	use Tot Shrink = 18%; keep the breakdown for the optimisation case	
10	WARN	structural break in the volume series	March: +87% MoM; all 5 languages move together (1.60x-2.06x)	treat as a level shift, not seasonality - anchor the baseline on post-break months only	determines the forecast baseline
11	FIX	SLA attainment series within [0,1]	1 month(s) report >100% attainment (February 110%)	exclude the impossible month from the attainment trend	reporting only - not a capacity input
12	WARN	SLA attainment trend	attainment moves 95% -> 92% (-3%) while volume grows - the current manual model is under-forecasting	use as the business case for the transformation	evidence, not an input

Grading matters. A ±2-contact rounding residue and a 110% SLA reading are both "wrong", but only one changes a decision. Every row above carries its own materiality, so the reader can see which findings were acted on and which were merely noted.

Forecast detail — why only English gets a trend



PER-LANGUAGE GROWTH TEST — POST-BREAK WINDOW

LANGUAGE	JUNE	MAR → JUN	OLS SLOPE /MONTH	R ²	Σ	GROWING?
English	158,298	+59.3%	+16,870	0.74	19.5%	yes
German	32,850	-15.7%	-1,904	0.68	6.9%	no
Spanish	29,890	+5.1%	+282	0.15	8.3%	no
French	24,238	-3.2%	-363	0.19	5.6%	no
Italian	21,878	+4.6%	-474	0.03	27.1%	no

A language earns the trend extrapolation in HIGH only if its slope is **positive**, exceeds 3% of baseline, and fits at $R^2 > 0.4$. Only English qualifies. An earlier version of this test keyed off the absolute slope, which flagged *declining* German as growing and produced a HIGH case below BASE — caught by the verification suite.

CONFIDENCE BAND — JULY

PCTILE	MULT	CONTACTS	PAID FTE
P5	0.797	234,264	405
P10	0.838	246,292	425
P25	0.911	267,784	463
P50	1.000	293,869	508
P75	1.097	322,496	557
P90	1.193	350,638	606
P95	1.254	368,641	637

Log-normal around the BASE median, $\sigma = 13.8\%$ from month-over-month log returns with the March break excluded. FTE is linear in volume, so the same multipliers carry through.

MONTHLY RE-FORECAST TRIGGER

SIGNAL	THRESHOLD	ACTION
English vs BASE	> +8%, 2 weeks	Switch to HIGH; release flex pool
English vs BASE	< -8%, 2 weeks	Switch to LOW; freeze pipeline
SLA attainment	< 90%, 2 weeks	Immediate re-forecast
AHT drift	±5% vs plan	Re-baseline AHT, not volume

The SLA trigger already fired — **May came in at 90%**. A model that only produces a number, without the rule for when the number is wrong, is not operational.

Why service level costs 34% — the Erlang mechanics

THE MISTAKE WORTH UNDERSTANDING

A workload model divides work by time and calls the result headcount. That is correct only if every agent is busy every second. **No queue with a service-level target can run that way** — to answer 80% of calls within 20 seconds, somebody has to be free when the call lands. The idle time is not waste; it is the product.

ENGLISH INBOUND, BASE JULY	
Contacts	44,182
AHT (BPO1/2 blend)	457.5 s
Workload hours	5,615
Offered load	7.69 erlangs
Seats for 80/20	11
Achieved service level	82.6%
Occupancy at 11 seats	69.9%
Workload FTE (+ shrinkage only)	39.5
FTE that actually meets SLA	56.5

+43% on one queue. Across all ten real-time queues it is +34% on the total — the gap between a plan that hits SLA and one that does not.

WHY SMALL QUEUES COST SO MUCH MORE

Occupancy is not a property of the agents — it is a property of the *queue size*. A large queue smooths its own arrivals; a small one cannot, so it must hold spare seats against randomness that a bigger pool would absorb.

QUEUE	ERLANGS	SEATS	OCCUPANCY
Italian inbound	0.84	3	27.9%
French inbound	0.93	3	30.9%
Spanish inbound	1.14	3	38.1%
German inbound	1.26	3	41.9%
All four, pooled	4.16	7	59.5%
English inbound	7.69	11	69.9%

This is the whole argument for Strategy 1, and it is invisible to a workload model — which would report these five queues as equally efficient per contact.

Doubling volume does not double FTE. The verification suite checks this directly: 2× the contacts needs **1.81×** the FTE. Scale is a real economy in this model, which is exactly why pooling pays.

WHERE THIS MODEL IS DELIBERATELY CONSERVATIVE

- Monthly-average arrival rates.** Erlang is run on a flat 24/7 rate. Real intraday peaks and overnight minimum-staffing floors make interval-level staffing *higher*, never lower. **508 is a floor.**
- Shrinkage held flat at 18%.** BPO 3 carries 7% vacation and Jul-Sep is peak holiday. A +1–2 point summer adjustment adds 6–12 FTE (see sensitivity below).
- Outbound kept at face value** despite mirroring inbound exactly. Removing it would cut ~8 FTE; keeping it is the conservative call.
- Email at 85% planned occupancy** — an assumption, not from the data.

SHRINKAGE SENSITIVITY

TOTAL SHRINKAGE	SLA FTE	Δ
16%	472.7	–11.5
17%	478.4	–5.8
18% — as given	484.2	—
19%	490.2	+6.0
20%	496.3	+12.1

Controllable buckets (break + lunch 6%, meetings 1%, coaching 1%) total **8 points**; vacation and sick are entitlement. "Cut shrinkage" is generic — **"reclaim 1–2 points from break scheduling and move coaching to low-demand hours"** is worth 6–12 FTE and is actionable this quarter.

Note the vendor difference the flat 18% hides: **BPO 3 carries 7% vacation and 3% sick; BPO 1 and 2 carry 5% each.** Same total, different labour model — and BPO 3's vacation line is the one that rises in the quarter being planned.

The model and how it was verified

THE MODEL

MODULE	RESPONSIBILITY
config.py	Every assumption in one frozen dataclass — nothing hard-coded downstream
erlang.py	Erlang B/C, service level, ASA, staffing solver, normal inverse CDF
loader.py	Schema-driven reader; refuses the hidden sheet; finds languages and channels by pattern, not cell address
clean.py	12 data-quality checks → structured log
forecast.py	Break detection, growth test, scenarios, confidence bands
capacity.py	Workload → Erlang → seats → FTE → paid headcount, by BPO
optimise.py	Each lever re-solved through the same engine
report.py	16-sheet Excel export

Automation was the point. Drop in next month's workbook and the plan regenerates: languages, channels and months are discovered from the sheet, and every assumption is a config value. The verification suite proves it by re-running with a language removed and with every constant altered.

```
python run_model.py
python run_model.py --aht-source reason
python crosscheck.py
```

VERIFICATION — 72 CHECKS, ALL PASSING

LAYER	WHAT IS ACTUALLY PROVED
Erlang B / C	Recursion vs the textbook closed form (max diff 1.1e-16); Erlang C vs the M/M/c stationary distribution derived independently (<1e-12)
Simulation	Erlang C vs a discrete-event M/M/c simulation — 5 replications × 300k arrivals per case; the analytic value sits inside a 3σ band on both service level and ASA
Solver	Returns the minimal feasible seat count; service level monotone in seats
Queueing	Little's Law $L_q = \lambda W_q$ holds exactly
Raw cells	The entire July chain re-derived from worksheet cells through a separate code path — matches to 1e-6
Invariants	Every queue achieves ≥80% (min 80.37%); queue FTE sums exactly to the roll-up; the BPO split re-aggregates with zero FTE lost; $LOW \leq BASE \leq HIGH$
HR maths	Attrition compounds to 36.00%/yr; tenure efficiency matches a 3,000-iteration cohort simulation
Levers	Each 1b strategy re-solved, not asserted — including the one that came back negative

WHAT VERIFICATION ACTUALLY CAUGHT

Three real defects **in my own model**, all fixed at source:

- The growth test keyed off **|slope|**, so *declining* German qualified as "growing" — and HIGH came out **below** BASE.
- The reported trend was a one-step projection (6,397/mo) rather than the OLS slope (**16,869/mo**).
- A fourth failure surfaced the ±2-contact residue between the channel rows and their own totals — which became data-quality finding #1 and forced an explicit decision about which row to trust.

One "failure" turned out to be wrong in the *test*, not the code: a reference value typed from memory. Four independent derivations — closed form, via Erlang B, the stationary distribution, and the simulation — agreed to **1e-12**. The test was corrected, not the model.

Why this belongs in the submission. A forecast is a number somebody staffs against. The useful question is not "is it right?" but "what would have to be true for it to be wrong, and did you check?" Every figure in this deck is generated by the model — the deck reads the model's own output, so the two cannot drift apart.

The honest limit. Verification proves the model computes what it claims. It cannot prove the inputs are right — and two of them are not yet settled. That is why the two open questions carry FTE price tags on slide 2 rather than being quietly resolved by assumption.

Assumptions register and open requests

EVERY ASSUMPTION, AND WHERE IT COMES FROM

ASSUMPTION	VALUE	NOTE	SOURCE
Hours per FTE per month	173.33	=40 × 52 ÷ 12	'Other information' B2
Total shrinkage	18%	divisor 0.82	'Shrinkage', identical all BPOs
Operating hours per month	730	24/7 → 8,760 ÷ 12	'Other information' B7
Phone SLA	80% ≤ 20 s	Erlang C	B8
Chat SLA	80% ≤ 60 s	Erlang C, ÷ concurrency	B9
Email SLA	80% ≤ 120 min	deferred model	B10
Chat concurrency	1.2	open question — worth 75 FTE	'AHT Assumptions' G column
Deferred occupancy	85%	modelling assumption — not in the data	—
Seasonal uplift	+10%	applied to the June baseline	the brief
Annual attrition	36%	compounds to 3.65%/month	B12
Learning curve	120 / 110 / 105%	AHT multiplier, months 1–3	B4
Training	4 weeks	non-producing, paid	B13
Routing	BPO 1&2 English; BPO 3 DE/IT/FR/ES	9 of 15 AHT rows unroutable	B5 / B6
English BPO split	50 / 50	AHT blended, so capacity is split-agnostic	assumption
English AHT source	channel-level (709.8 s)	vs reason-level 908.8 s — worth 78 FTE	'AHT Assumptions'

WHAT I WOULD WANT BEFORE THE NEXT CYCLE

- **Which English AHT is authoritative.** Worth **78 FTE**. Needs an owner, not more analysis.
- **Whether chat concurrency was measured.** Worth **75 FTE**.
- **24+ months of history.** Six months containing a structural break cannot support a fitted seasonal model. Until then the forecast is scenario-based by necessity, not by preference.
- **15/30-minute arrival profiles.** Monthly-average Erlang is a floor; intraday shape is where the real staffing lives.
- **What happened in March.** A uniform 1.87× step across five languages, four channels and 49 reason codes is a system or scope change, not demand.
- **Cost per productive hour by BPO.** Contracts are priced this way, and without it no lever can be ranked in €.
- **True outbound volumes.** The mirrored columns cannot be real.

How I would use the team against this forecast

One specialist owns English — the volume, the growth and the volatility all sit there. One owns the pooled non-English queues, which is where Strategy 1 gets implemented. One owns the data contract and the re-forecast trigger, so the assumptions have a named keeper rather than living in a spreadsheet nobody edits. I own the two open questions, because they need a decision from the business rather than an analysis.

Deliverables: this deck · the source workbook with working shown · a 16-sheet model output workbook · the Python model and its 72-check verification suite.